Name: _____

# Speed Challenge



**Step 1: Gather your materials**
Each team needs:

- 1 stopwatch
- 1 metre ruler or measuring tape
- 3 markers or cones

**Step 2: Create your race track**
Use the metre ruler to measure a 10 m track, marking the halfway point at 5 m.

**Step 3: Record your data**
Each team member will need to perform the following tasks at each distance: walking normally, walking backwards and hopping.

To improve the accuracy of your results you will need to repeat each trial three times and calculate an average.

| Task | Distance | Time Trial 1 | Time Trial 2 | Time Trial 3 | Average Time |
|---|---|---|---|---|---|
| **Walking Normally** | 5 m | | | | |
| | 10 m | | | | |
| **Walking Backwards** | 5 m | | | | |
| | 10 m | | | | |
| **Hopping** | 5 m | | | | |
| | 10 m | | | | |

**Step 4: Calculate your average speed for each activity**

$$\textbf{Speed} = \frac{\textbf{Distance}}{\textbf{Time}}$$

Speed Walking Normally (5 m) = _____ =

Speed Walking Normally (10 m) = _____ =

Speed Walking backwards (5 m) = _____ =

Speed Walking backwards (10 m) = _____ =

Speed Hopping (5 m) = _____ =

Speed Hopping (10 m) = _____ =

**Step 4: Questions**

1. Which task and distance resulted in your fastest speed?

   Task = _____ Distance = _____ Speed = _____

2. Which task and distance resulted in the slowest speed?

   Task = _____ Distance = _____ Speed = _____

3. Rearrange the speed equation to the find distance.

4. Rearrange the speed equation to find the time.

5. How far could you walk normally in 10 minutes based on your average speed for the 5m trials?

6. How long would it take you to hop 30 metres based on your average speed for the 5m trials?

7. How far could you travel walking backwards for 15 minutes based on your average speed for the 10m trials?

8. Based on your average walking speed from the 10m trials, how long would it take you to walk 1km?

9. Based on your average hopping speed from the 10m trials, how long would it take you to hop 1km?

10. Do you think your results are accurate? Why or why not? How could you improve your accuracy?